

SQL Benchmark Submission Report

Database: classicmodels (from week2 seed)
Report Overview: This report documents the SQL benchmark submission for Part 1 and Part 2. Part 1 contains SQL query implementations and execution screenshots. Part 2 presents the evaluation strategy for Text-to-SQL agents, including execution accuracy, robustness, safety, and efficiency metrics.

Part 1: SQL Query Tasks

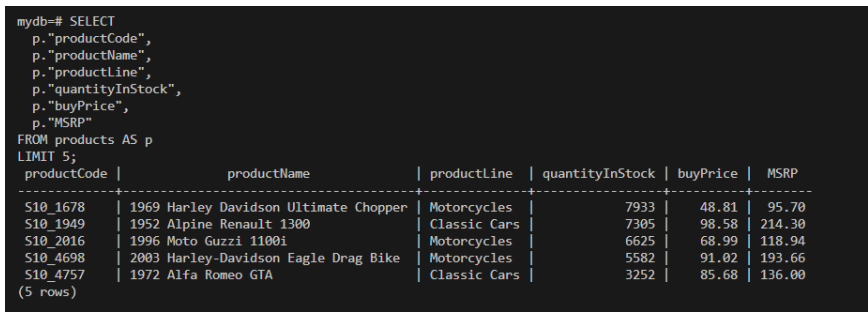
Part 1 includes a comprehensive collection of SQL queries executed against the classicmodels database. The tasks covered:

- Basic SELECT queries
- DISTINCT queries
- JOIN operations
- Aggregation and GROUP BY queries
- Aggregate functions such as COUNT, SUM, AVG, MIN, and MAX
- Customer, employee, payment, product, and order analysis

A total of 50 SQL tasks were completed and validated using PostgreSQL. Execution screenshots from the completed tasks are included below.

Execution Screenshots

Figure 1: SQL Query Execution Result

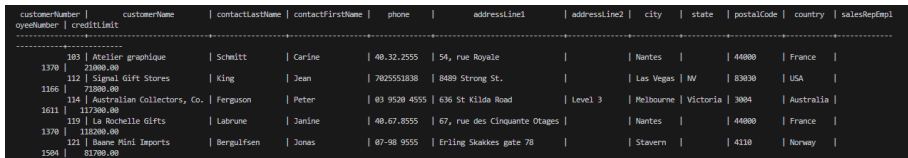


```
mydb=# SELECT
p."productCode",
p."productName",
p."productline",
p."quantityInStock",
p."buyPrice",
p."MSRP"
FROM products AS p
LIMIT 5;
```

productCode	productName	productline	quantityInStock	buyPrice	MSRP
S10_1678	1969 Harley Davidson Ultimate Chopper	Motorcycles	7933	48.81	95.70
S10_1949	1952 Alpine Renault 1300	Classic Cars	7305	98.58	214.30
S10_2016	1996 Moto Guzzi 1100i	Motorcycles	6625	68.99	118.94
S10_4698	2003 Harley-Davidson Eagle Drag Bike	Motorcycles	5582	91.02	193.66
S10_4757	1972 Alfa Romeo GTA	Classic Cars	3252	85.68	136.00

(5 rows)

Figure 2: SQL Query Execution Result



customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepName
1370	Atelier graphique	Schmitt	Carine	40-32-2555	54, rue Royale		Nantes		44000	France	
1166	Signal Gift Stores	King	Jean	7025551038	8489 Strong St.		Las Vegas	NV	83030	USA	
114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	436 St Klida Road	Level 3	Melbourne	Victoria	3004	Australia	
1011	La Rochelle Gifts	Labrun	Jarline	40-67-8555	67, rue des Cinquante Otages		Nantes		44000	France	
121	Boone Mini Imports	Bergulfsen	Jones	07-98 9555	Eriling Skakkes gate 78		Stavern		4110	Norway	

Figure 3: SQL Query Execution Result

orderNumber	orderDate	requiredDate	shippedDate	status	customerNumber	comments
10180	2003-01-06	2003-01-13	2003-01-10	Shipped	363	
10181	2003-01-09	2003-01-18	2003-01-11	Shipped	128	Check on availability.
10182	2003-01-10	2003-01-18	2003-01-14	Shipped	181	
10183	2003-01-29	2003-02-07	2003-02-02	Shipped	121	
10184	2003-01-31	2003-02-09	2003-02-01	Shipped	141	
10185	2003-02-11	2003-02-21	2003-02-12	Shipped	145	
10186	2003-02-17	2003-02-24	2003-02-21	Shipped	278	
10187	2003-02-24	2003-03-03	2003-02-26	Shipped	131	Difficult to negotiate with customer. We need more marketing materials
10188	2003-03-03	2003-03-12	2003-03-08	Shipped	385	
10189	2003-03-10	2003-03-19	2003-03-11	Shipped	486	Customer requested that FedEx Ground is used for this shipping
10110	2003-03-18	2003-03-24	2003-03-20	Shipped	187	
10111	2003-03-25	2003-03-31	2003-03-30	Shipped		

Figure 4: SQL Query Execution Result

```
mydb=# SELECT
e."employeeNumber",
e."lastName",
e."firstName",
e."extension",
e."email",
e."officeCode",
e."reportsTo",
e."jobTitle"
FROM employees AS e;
```

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1		President
1056	Patterson	Mary	x4611	mpatterson@classicmodelcars.com	1	1002	VP Sales
1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing
1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep
1166	Thompson	Leslie	x4965	lthompson@classicmodelcars.com	1	1143	Sales Rep
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep
1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep
1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep
1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102	Sales Rep
1370	Hernandez	Gerard	x2028	ghernandez@classicmodelcars.com	4	1102	Sales Rep
1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1102	Sales Rep
1501	Bott	Larry	x2311	lbott@classicmodelcars.com	7	1102	Sales Rep
1504	Jones	Barry	x102	bjones@classicmodelcars.com	7	1102	Sales Rep
1611	Fixter	Andy	x101	afixter@classicmodelcars.com	6	1088	Sales Rep
1612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	6	1088	Sales Rep
1619	King	Tom	x103	tking@classicmodelcars.com	6	1088	Sales Rep
1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1056	Sales Rep
1625	Kato	Yoshimi	x102	ykato@classicmodelcars.com	5	1621	Sales Rep
1702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	4	1102	Sales Rep

(23 rows)

Figure 5: SQL Query Execution Result

```
mydb=# SELECT
o."officeCode",
o."city",
o."phone",
o."addressLine1",
o."addressLine2",
o."state",
o."country",
o."postalCode",
o."territory"
FROM offices AS o;
```

officeCode	city	phone	addressLine1	addressLine2	state	country	postalCode	territory
1	San Francisco	+1 650 219 4782	100 Market Street	Suite 300	CA	USA	94080	NA
2	Boston	+1 215 837 8825	1550 Court Place	Suite 102	MA	USA	02107	NA
3	NYC	+1 212 555 3000	523 East 53rd Street	apt. 5A	NY	USA	10022	NA
4	Paris	+33 14 723 4484	43 Rue Jouffroy Dabbans			France	75017	EMEA
5	Tokyo	+81 33 224 5000	4-1 Kioicho		Chiyoda-Ku	Japan	102-8578	Japan
6	Sydney	+61 2 9264 2451	5-11 Wentworth Avenue	Floor #2		Australia	NSW 2010	APAC
7	London	+44 20 7877 2841	25 Old Broad Street	Level 7		UK	EC2N 1HN	EMEA

(7 rows)

Figure 6: SQL Query Execution Result

productLine	vehicleDescription	materialDescription	image
Classic Cars	Classic Cars - A museum car enthusiasts take your classic car ownership down one true, whether you are looking for classic muscle cars, three-wheeled cars or more unusual recreations, you will find great choices in this category. These replicas feature expert attention to detail and top end features such as working steering system, working engine components, opening rear door and removable hard shell, a steel independent spring suspension, and so on. The models range in size from 1:18 to 1:24 scale and include numerous limited edition and several sets of products. Replicas - The replicas are state of the art replicas of classic as well as contemporary automobiles (sports such as Harley Davidson, Ducati and Honda. Models contain steering details such as official logos, rotating wheels, working handbrake, front suspension, gear shift lever, fuel and water tanks, mirrors and engine hood and so on. The models range in size from 1:18 to 1:24 scale and include numerous limited edition and several sets of products. Replicas - The replicas are state of the art replicas of classic as well as contemporary automobiles (sports such as Harley Davidson, Ducati and Honda. Models contain steering details such as official logos and handbrake, rotating front suspension and mirrors, and so on. Next come in size and with a certificate of authenticity from their manufacturers.		
Model Cars	Model Cars - The perfect holiday or anniversary gift for executives, clients, friends, and family. These handcrafted model cars are unique, stunning works of art that will be treasured for generations! They come fully assembled and ready for display in the home or office. No generation, quality, and best value.		
Trains	Trains - Model trains are a wonderful hobby for enthusiasts of all ages, whether you are looking for collectible model trains, electric locomotives or locomotives, you will find a number of great choices for any budget within this category. The interactive aspect of trains makes toy and the young children. The model trains are also ideal for children under the age of 5.		
Boats and Boats	Boats and Boats - The boats and the boats are realistic replicas of boats and specialized boats produced from the early 1900s to present. The models range in size from 1:18 to 1:24 scale and include numerous limited edition and several sets of products. Materials used include 1:18 and 1:24 scale. All models include a certificate of authenticity from their manufacturers and are a perfect addition for the home or office.		
Motor Cars	Motor Cars - The motor cars are realistic replicas of automobiles produced from the early 1900s to present. The models range in size from 1:18 to 1:24 scale and include numerous limited edition and several sets of products. Materials used include 1:18 and 1:24 scale. All models include a certificate of authenticity from their manufacturers and are a perfect addition for the home or office.		
Motor Cars	Motor Cars - The motor cars are realistic replicas of automobiles produced from the early 1900s to present. The models range in size from 1:18 to 1:24 scale and include numerous limited edition and several sets of products. Materials used include 1:18 and 1:24 scale. All models include a certificate of authenticity from their manufacturers and are a perfect addition for the home or office.		

Figure 7: SQL Query Execution Result

customerNumber	checkNumber	paymentDate	amount
103	HQ336336	2004-10-19	6066.78
103	JM555205	2003-06-05	14571.44
103	OM314933	2004-12-18	1676.14
112	B0864823	2004-12-17	14191.12
112	HQ55022	2003-06-06	32641.98
112	ND748579	2004-08-20	33347.88
114	GG31455	2003-05-20	45864.03
114	MA765515	2004-12-15	82261.22
114	NP603840	2003-05-31	7565.08
114	NR27552	2004-03-10	44894.74
119	DB933704	2004-11-14	19501.82
119	LN373447	2004-08-08	47924.19
119	NG94694	2005-02-22	49523.67
121	DB889831	2003-02-16	50218.95
121	FD317790	2003-10-28	1491.38
121	KI8311359	2004-11-04	17876.32
121	MA302151	2004-11-28	34638.14
124	AE215433	2005-03-05	101244.59
124	BG255406	2004-08-28	85410.87
124	CQ287967	2003-04-11	11044.30
124	ET64396	2005-04-16	83598.04
124	HI366474	2004-12-27	47142.70

Figure 8: SQL Query Execution Result

productName	buyPrice	MSRP
1969 Harley Davidson Ultimate Chopper	48.81	95.70
1952 Alpine Renault 1300	98.58	214.30
1996 Moto Guzzi 1100i	68.99	118.94
2003 Harley-Davidson Eagle Drag Bike	91.02	193.66
1972 Alfa Romeo GTA	85.68	136.00
1962 Lancia Delta 16V	103.42	147.74
1968 Ford Mustang	95.34	194.57
2001 Ferrari Enzo	95.59	207.80
1958 Setra Bus	77.90	136.67
2002 Suzuki XRE0	66.27	150.62
1969 Corvair Monza	89.14	151.08
1968 Dodge Charger	75.16	117.44
1969 Ford Falcon	83.05	173.02
1970 Plymouth Hemi Cuda	31.92	79.80
1957 Chevy Pickup	55.70	118.50
1969 Dodge Charger	58.73	115.16
1940 Ford Pickup Truck	58.33	116.67
1993 Mazda RX-7	83.51	141.54
1937 Lincoln Berline	60.62	102.74
1936 Mercedes-Benz 500K Special Roadster	24.26	53.91
1965 Aston Martin DB5	65.96	124.44
1980s Black Hawk Helicopter	77.27	157.69
1917 Grand Touring Sedan	86.70	170.00
1948 Porsche 356-A Roadster	53.90	77.00

Figure 9: SQL Query Execution Result

customerName	city
Atelier graphique	Nantes
Signal Gift Stores	Las Vegas
Australian Collectors, Co.	Melbourne
La Rochelle Gifts	Nantes
Baane Mini Imports	Stavern
Mini Gifts Distributors Ltd.	San Rafael
Havel & Zbyszek Co	Warszawa
Blauer See Auto, Co.	Frankfurt
Mini Wheels Co.	San Francisco
Land of Toys Inc.	NYC
Euro+ Shopping Channel	Madrid
Volvo Model Replicas, Co	Lule??
Danish Wholesale Imports	Kobenhavn
Saveley & Henriot, Co.	Lyon
Dragon Souvenirs, Ltd.	Singapore
Muscle Machine Inc	NYC
Diecast Classics Inc.	Allentown
Technics Stores Inc.	Burlingame
Handji Gifts& Co	Singapore
Herkku Gifts	Bergen
American Souvenirs Inc	New Haven

Figure 10: SQL Query Execution Result

```
mydb=# SELECT
e."firstName",
e."lastName"
FROM employees AS e;
 firstName | lastName
-----+-----
 Diane    | Murphy
 Mary     | Patterson
 Jeff     | Firrelli
 William  | Patterson
 Gerard   | Bondur
 Anthony  | Bow
 Leslie   | Jennings
 Leslie   | Thompson
 Julie    | Firrelli
 Steve    | Patterson
 Foon Yue | Tseng
 George   | Vanauf
 Loui     | Bondur
 Gerard   | Hernandez
 Pamela   | Castillo
 Larry    | Bott
 Barry    | Jones
 Andy     | Fixter
 Peter    | Marsh
 Tom      | King
 Mami     | Nishi
 Yoshimi  | Kato
 Martin   | Gerard
(23 rows)
```

Figure 11: SQL Query Execution Result

```
orderDate
-----
2003-01-06
2003-01-09
2003-01-10
2003-01-29
2003-01-31
2003-02-11
2003-02-17
2003-02-24
2003-03-03
2003-03-10
2003-03-18
2003-03-25
2003-03-24
2003-03-26
2003-04-01
2003-04-04
2003-04-11
2003-04-16
2003-04-21
2003-04-28
2003-04-29
2003-05-07
2003-05-08
2003-05-20
2003-05-21
2003-05-21
2003-05-28
2003-06-03
2003-06-06
2003-06-12
2003-06-16
2003-06-16
2003-06-25
2003-06-27
```

Figure 12: SQL Query Execution Result

```
mydb=# SELECT DISTINCT
p."productVendor"
FROM products AS p
ORDER BY p."productVendor";
 productVendor
-----
Autoart Studio Design
Carousel DieCast Legends
Classic Metal Creations
Exoto Designs
Gearbox Collectibles
Highway 66 Mini Classics
Min Lin Diecast
Motor City Art Classics
Red Start Diecast
Second Gear Diecast
Studio M Art Models
Unimax Art Galleries
Welly Diecast Productions
(13 rows)
```

Figure 13: SQL Query Execution Result

```
productCode
-----
S10_1678
S10_1949
S10_2016
S10_4698
S10_4757
S10_4962
S12_1099
S12_1108
S12_1666
S12_2823
S12_3148
S12_3380
S12_3891
S12_3990
S12_4473
S12_4675
S18_1097
S18_1129
S18_1342
S18_1367
S18_1589
S18_1662
S18_1749
S18_1889
```

Figure 14: SQL Query Execution Result

```
mydb=# SELECT DISTINCT
o."country"
FROM offices AS o
ORDER BY o."country";
country
-----
Australia
France
Japan
UK
USA
(5 rows)
```

Figure 15: SQL Query Execution Result

```
mydb=# SELECT DISTINCT
o."status"
FROM orders AS o
ORDER BY o."status";
status
-----
Cancelled
Disputed
In Process
On Hold
Resolved
Shipped
(6 rows)
```

Figure 16: SQL Query Execution Result

```
amount
-----
6066.78
14571.44
1676.14
14191.12
32641.98
33347.88
45864.03
82261.22
7565.08
44894.74
19501.82
47924.19
49523.67
50218.95
1491.38
17876.32
34638.14
101244.59
85410.87
11044.30
83598.04
47142.70
55639.66
```

Figure 17: SQL Query Execution Result

```
mydb=# SELECT DISTINCT
e."jobTitle"
FROM employees AS e
ORDER BY e."jobTitle";
jobTitle
-----
President
Sale Manager (EMEA)
Sales Manager (APAC)
Sales Manager (NA)
Sales Rep
VP Marketing
VP Sales
(7 rows)
```

Figure 18: SQL Query Execution Result

customerName	phone
Atelier graphique	40.32.2555
Signal Gift Stores	7025551838
Australian Collectors, Co.	03 9520 4555
La Rochelle Gifts	40.67.8555
Baane Mini Imports	07-98 9555
Mini Gifts Distributors Ltd.	4155551450
Havel & Zbyszek Co	(26) 642-7555
Blauer See Auto, Co.	+49 69 66 90 255
Mini Wheels Co.	6505555787
Land of Toys Inc.	2125557818
Euro+ Shopping Channel	(91) 555 94 44
Volvo Model Replicas, Co	0921-12 3555
Danish Wholesale Imports	31 12 3555
Saveley & Henriot, Co.	78.32.5555
Dragon Souveniers, Ltd.	+65 221 7555
Muscle Machine Inc	2125557413
Diecast Classics Inc.	2155551555
Technics Stores Inc.	6505556809

Figure 19: SQL Query Execution Result

productName	MSRP
1969 Harley Davidson Ultimate Chopper	95.70
1952 Alpine Renault 1300	214.30
1996 Moto Guzzi 1100i	118.94
2003 Harley-Davidson Eagle Drag Bike	193.66
1972 Alfa Romeo GTA	136.00
1962 LanciaA Delta 16V	147.74
1968 Ford Mustang	194.57
2001 Ferrari Enzo	207.80
1958 Setra Bus	136.67
2002 Suzuki XRE0	150.62
1969 Corvair Monza	151.08
1968 Dodge Charger	117.44
1969 Ford Falcon	173.02
1970 Plymouth Hemi Cuda	79.80
1957 Chevy Pickup	118.50
1969 Dodge Charger	115.16
1940 Ford Pickup Truck	116.67
1993 Mazda RX-7	141.54
1937 Lincoln Berline	102.74
1936 Mercedes-Benz 500K Special Roadster	53.91
1965 Aston Martin DB5	124.44
1980s Black Hawk Helicopter	157.69
1917 Grand Touring Sedan	170.00
1948 Porsche 356-A Roadster	77.00
1995 Honda Civic	142.25

Figure 20: SQL Query Execution Result

orderNumber
10100
10101
10102
10103
10104
10105
10106
10107
10108
10109
10110
10111
10112
10113
10114
10115
10116
10117
10118
10119
10120
10121
10122

Figure 21: SQL Query Execution Result

orderNumber	orderDate	status	customerName
10100	2003-01-06	Shipped	Online Diecast Creations Co.
10101	2003-01-09	Shipped	Blauer See Auto, Co.
10102	2003-01-10	Shipped	Vitachrome Inc.
10103	2003-01-29	Shipped	Baane Mini Imports
10104	2003-01-31	Shipped	Euro+ Shopping Channel
10105	2003-02-11	Shipped	Danish Wholesale Imports
10106	2003-02-17	Shipped	Rovelli Gifts
10107	2003-02-24	Shipped	Land of Toys Inc.
10108	2003-03-03	Shipped	Cruz & Sons Co.
10109	2003-03-10	Shipped	Motor Mint Distributors Inc.
10110	2003-03-18	Shipped	AV Stores, Co.
10111	2003-03-25	Shipped	Mini Wheels Co.
10112	2003-03-24	Shipped	Volvo Model Replicas, Co
10113	2003-03-26	Shipped	Mini Gifts Distributors Ltd.
10114	2003-04-01	Shipped	La Corne Dabondance, Co.
10115	2003-04-04	Shipped	Classic Legends Inc.
10116	2003-04-11	Shipped	Royale Belge
10117	2003-04-16	Shipped	Dragon Souveniers, Ltd.
10118	2003-04-21	Shipped	Enaco Distributors
10119	2003-04-28	Shipped	Salzburg Collectables
10120	2003-04-29	Shipped	Australian Collectors, Co.
10121	2003-05-07	Shipped	Reims Collectables
10122	2003-05-08	Shipped	Marseille Mini Autos
10123	2003-05-20	Shipped	Atelier graphique
10124	2003-05-21	Shipped	Signal Gift Stores

Figure 22: SQL Query Execution Result

```
mysql> SELECT
  e."employeeNumber",
  e."firstName",
  e."lastName",
  o."city" AS "officeCity"
FROM employees AS e
JOIN offices AS o
ON o."officeCode" = e."officeCode";
```

employeeNumber	firstName	lastName	officeCity
1002	Diane	Murphy	San Francisco
1005	Mary	Patterson	San Francisco
1006	Jeff	Fairrelli	San Francisco
1008	William	Patterson	Sydney
1102	Gerard	Bondur	Paris
1143	Anthony	Bow	San Francisco
1165	Leslie	Jennings	San Francisco
1166	Leslie	Thompson	San Francisco
1188	Julie	Fairrelli	Boston
1216	Steve	Patterson	Boston
1206	Foon Yue	Tsang	NYC
1323	George	Vanauf	NYC
1337	Loui	Bondur	Paris
1370	Gerard	Hernandez	Paris
1401	Pamela	Castillo	Paris
1501	Larry	Bett	London
1504	Barry	Jones	London
1611	Andy	Fixter	Sydney
1612	Peter	Marsh	Sydney
1619	Tom	King	Sydney
1621	Hani	Nishi	Tokyo
1625	Yoshimi	Kato	Tokyo
1702	Martin	Gerard	Paris

(23 rows)

Figure 23: SQL Query Execution Result

customerNumber	customerName	checkNumber	paymentDate	amount
103	Atelier graphique	HQ336336	2004-10-19	6066.78
103	Atelier graphique	JW555205	2003-06-05	14571.44
103	Atelier graphique	OW314933	2004-12-18	1676.14
112	Signal Gift Stores	B0864823	2004-12-17	14191.12
112	Signal Gift Stores	HQ55022	2003-06-06	32641.98
112	Signal Gift Stores	ND748579	2004-08-20	33347.88
114	Australian Collectors, Co.	GG31455	2003-05-20	45864.03
114	Australian Collectors, Co.	MA765515	2004-12-15	82261.22
114	Australian Collectors, Co.	NP603840	2003-05-31	7565.08
114	Australian Collectors, Co.	NR27552	2004-03-10	44894.74
119	La Rochelle Gifts	DB933704	2004-11-14	19501.82
119	La Rochelle Gifts	LN373447	2004-08-08	47924.19
119	La Rochelle Gifts	NG94694	2005-02-22	49523.67
121	Baane Mini Imports	DB889831	2003-02-16	50218.95
121	Baane Mini Imports	FD317790	2003-10-28	1491.38
121	Baane Mini Imports	KI831359	2004-11-04	17876.32
121	Baane Mini Imports	MA302151	2004-11-28	34638.14
124	Mini Gifts Distributors Ltd.	AE215433	2005-03-05	101244.59
124	Mini Gifts Distributors Ltd.	BG255406	2004-08-28	85410.87
124	Mini Gifts Distributors Ltd.	CQ287967	2003-04-11	11044.30
124	Mini Gifts Distributors Ltd.	ET64396	2005-04-16	83598.04
124	Mini Gifts Distributors Ltd.	HI366474	2004-12-27	47142.70
124	Mini Gifts Distributors Ltd.	HR86578	2004-11-02	55639.66
124	Mini Gifts Distributors Ltd.	KI131716	2003-08-15	111654.40
124	Mini Gifts Distributors Ltd.	LF217299	2004-03-26	43369.30
124	Mini Gifts Distributors Ltd.	NT141748	2003-11-25	45084.38
128	Blauer See Auto, Co.	DT925118	2003-01-28	10549.01
128	Blauer See Auto, Co.	FA465482	2003-10-18	24101.81
128	Blauer See Auto, Co.	FH668230	2004-03-24	33820.62

Figure 24: SQL Query Execution Result

orderNumber	productCode	productName	quantityOrdered	priceEach	orderLineNumber
10100	S18_1749	1917 Grand Touring Sedan	30	136.00	3
10100	S18_2248	1911 Ford Town Car	50	55.09	2
10100	S18_4409	1932 Alfa Romeo 8C2300 Spider Sport	22	75.46	4
10100	S24_3969	1936 Mercedes-Benz 500k Roadster	49	35.29	1
10101	S18_2325	1932 Model A Ford 3-Coupe	25	100.06	4
10101	S18_2795	1920 Mercedes-Benz SSK	26	167.06	1
10101	S24_1937	1939 Chevrolet Deluxe Coupe	45	32.53	3
10101	S24_2022	1938 Cadillac V-16 Presidential Limousine	46	44.35	2
10102	S18_1342	1937 Lincoln Berline	39	95.55	2
10102	S18_1367	1936 Mercedes-Benz 500K Special Roadster	41	43.13	1
10103	S10_1949	1952 Alpine Renault 1300	26	214.30	11
10103	S10_4962	1962 Lancia Delta 16V	42	119.67	4
10103	S12_1666	1958 Setra Bus	27	121.64	8
10103	S18_1897	1940 Ford Pickup Truck	35	94.50	10
10103	S10_2432	1926 Ford Fire Engine	22	50.34	2
10103	S18_2049	1913 Ford Model T Speedster	27	92.19	12
10103	S18_2957	1934 Ford V8 Coupe	35	61.84	14
10103	S18_3136	18th Century Vintage Horse Carriage	25	86.92	13
10103	S18_3320	1917 Maxwell Touring Car	46	86.31	16
10103	S18_4600	1940s Ford truck	36	98.07	5
10103	S18_4668	1939 Cadillac Limousine	41	40.75	9
10103	S24_2300	1962 Volkswagen Microbus	36	107.34	1
10103	S24_4258	1936 Chrysler Airflow	25	88.62	15
10103	S32_1268	1980's GM Manhattan Express	31	92.46	3
10103	S32_3522	1996 Peterbilt 379 Stake Bed with Outrigger	45	63.35	7
10103	S700_2024	1902 Camaro Z28	42	94.07	6

Figure 25: SQL Query Execution Result

customerNumber	customerName	salesRepEmployeeNumber	salesRepFirstName	salesRepLastName
103	Atelier graphique	1370	Gerard	Hernandez
112	Signal Gift Stores	1166	Leslie	Thompson
114	Australian Collectors, Co.	1611	Andy	Fixter
119	La Rochelle Gifts	1370	Gerard	Hernandez
121	Baane Mini Imports	1504	Barry	Jones
124	Mini Gifts Distributors Ltd.	1165	Leslie	Jennings
125	Havel & Zbyszek Co.			
128	Blauer See Auto, Co.	1504	Barry	Jones
129	Mini Wheels Co.	1165	Leslie	Jennings
131	Land of Toys Inc.	1323	George	Vanauf
141	Euro+ Shopping Channel	1370	Gerard	Hernandez
144	Volvo Model Replicas, Co	1504	Barry	Jones
145	Danish Wholesale Imports	1401	Pamela	Castillo
146	Saveley & Henriot, Co.	1337	Loui	Bondur
148	Dragon Souvenirs, Ltd.	1621	Mami	Nishi
151	Muscle Machine Inc.	1286	Foon Yue	Tseng
157	Diecast Classics Inc.	1216	Steve	Patterson
161	Technics Stores Inc.	1165	Leslie	Jennings
166	Handji Gifts Co	1612	Peter	Marsh
167	Hankku Gifts	1504	Barry	Jones
168	American Souvenirs Inc	1286	Foon Yue	Tseng
169	Porto Imports Co.			

Figure 26: SQL Query Execution Result

orderNumber	orderDate	status	customerName	city
10100	2003-01-06	Shipped	Online Diecast Creations Co.	Nashua
10101	2003-01-09	Shipped	Blauer See Auto, Co.	Frankfurt
10102	2003-01-10	Shipped	Vitachrome Inc.	NYC
10103	2003-01-29	Shipped	Baane Mini Imports	Stavern
10104	2003-01-31	Shipped	Euro+ Shopping Channel	Madrid
10105	2003-02-11	Shipped	Danish Wholesale Imports	Kobenhavn
10106	2003-02-17	Shipped	Rovelli Gifts	Bergamo
10107	2003-02-24	Shipped	Land of Toys Inc.	NYC
10108	2003-03-03	Shipped	Cruz & Sons Co.	Makati City
10109	2003-03-10	Shipped	Motor Mint Distributors Inc.	Philadelphia
10110	2003-03-18	Shipped	AV Stores, Co.	Manchester
10111	2003-03-25	Shipped	Mini Wheels Co.	San Francisco
10112	2003-03-24	Shipped	Volvo Model Replicas, Co	Lule??
10113	2003-03-26	Shipped	Mini Gifts Distributors Ltd.	San Rafael
10114	2003-04-01	Shipped	La Corne Dabondance, Co.	Paris
10115	2003-04-04	Shipped	Classic Legends Inc.	NYC
10116	2003-04-11	Shipped	Royale Belge	Charleroi
10117	2003-04-16	Shipped	Dragon Souvenirs, Ltd.	Singapore
10118	2003-04-21	Shipped	Enaco Distributors	Barcelona
10119	2003-04-28	Shipped	Salzburg Collectables	Salzburg
10120	2003-04-29	Shipped	Australian Collectors, Co.	Melbourne
10121	2003-05-07	Shipped	Reims Collectables	Reims
10122	2003-05-08	Shipped	Marseille Mini Autos	Marseille
10123	2003-05-20	Shipped	Atelier graphique	Nantes
10124	2003-05-21	Shipped	Signal Gift Stores	Las Vegas
10125	2003-05-21	Shipped	Australian Collectors, Co.	Melbourne
10126	2003-05-28	Shipped	Corrida Auto Replicas, Ltd	Madrid
10127	2003-06-03	Shipped	Muscle Machine Inc	NYC

Figure 27: SQL Query Execution Result

```
mysql> SELECT
e."employeeNumber",
e."firstName",
e."lastName",
m."employeeNumber" AS "managerNumber",
m."firstName" AS "managerFirstName",
m."lastName" AS "managerLastName"
FROM employees AS e
LEFT JOIN employees AS m
ON m."employeeNumber" = e."reportsTo";
```

employeeNumber	firstName	lastName	managerNumber	managerFirstName	managerLastName
1002	Diane	Murphy			
1056	Mary	Patterson	1002	Diane	Murphy
1076	Jeff	Firelli	1002	Diane	Murphy
1088	William	Patterson	1056	Mary	Patterson
1102	Gerard	Bondur	1056	Mary	Patterson
1143	Anthony	Bow	1056	Mary	Patterson
1165	Leslie	Jennings	1143	Anthony	Bow
1166	Leslie	Thompson	1143	Anthony	Bow
1188	Julie	Firelli	1143	Anthony	Bow
1216	Steve	Patterson	1143	Anthony	Bow
1286	Foon Yue	Tseung	1143	Anthony	Bow
1323	George	Vanauf	1143	Anthony	Bow
1337	Loui	Bondur	1102	Gerard	Bondur
1370	Gerard	Hernandez	1102	Gerard	Bondur
1401	Pamela	Castillo	1102	Gerard	Bondur
1501	Larry	Bott	1102	Gerard	Bondur
1504	Barry	Jones	1102	Gerard	Bondur
1611	Andy	Fixter	1088	William	Patterson
1612	Peter	Marsh	1088	William	Patterson
1619	Tom	King	1088	William	Patterson
1621	Mami	Nishi	1056	Mary	Patterson
1625	Yoshimi	Kato	1621	Mami	Nishi
1702	Martin	Gerard	1102	Gerard	Bondur

(23 rows)

Figure 28: SQL Query Execution Result

orderNumber	productCode	productVendor	quantityOrdered	priceEach
10100	S18_1749	Welly Diecast Productions	30	136.00
10100	S18_2248	Motor City Art Classics	50	55.09
10100	S18_4409	Exoto Designs	22	75.46
10100	S24_3969	Red Start Diecast	49	35.29
10101	S18_2325	Autoart Studio Design	25	108.06
10101	S18_2795	Gearbox Collectibles	26	167.06
10101	S24_1937	Motor City Art Classics	45	32.53
10101	S24_2022	Classic Metal Creations	46	44.35
10102	S18_1342	Motor City Art Classics	39	95.55
10102	S18_1367	Studio M Art Models	41	43.13
10103	S10_1949	Classic Metal Creations	26	214.30
10103	S10_4962	Second Gear Diecast	42	119.67
10103	S12_1666	Welly Diecast Productions	27	121.64
10103	S18_1097	Studio M Art Models	35	94.50
10103	S18_2432	Carousel DieCast Legends	22	58.34
10103	S18_2949	Carousel DieCast Legends	27	92.19
10103	S18_2957	Min Lin Diecast	35	61.84
10103	S18_3136	Red Start Diecast	25	86.92
10103	S18_3320	Exoto Designs	46	86.31
10103	S18_4600	Motor City Art Classics	36	98.07
10103	S18_4668	Studio M Art Models	41	40.75
10103	S24_2300	Autoart Studio Design	36	107.34
10103	S24_4258	Second Gear Diecast	25	88.62
10103	S32_1268	Motor City Art Classics	31	92.46
10103	S32_3522	Red Start Diecast	45	63.35
10103	S700_2824	Carousel DieCast Legends	42	94.07
10104	S12_3148	Welly Diecast Productions	34	131.44

Figure 29: SQL Query Execution Result

customerNumber	customerName	country	paymentDate	amount
103	Atelier graphique	France	2004-10-19	6066.78
103	Atelier graphique	France	2003-06-05	14571.44
103	Atelier graphique	France	2004-12-18	1676.14
112	Signal Gift Stores	USA	2004-12-17	14191.12
112	Signal Gift Stores	USA	2003-06-06	32641.98
112	Signal Gift Stores	USA	2004-08-20	33347.88
114	Australian Collectors, Co.	Australia	2003-05-20	45864.03
114	Australian Collectors, Co.	Australia	2004-12-15	82261.22
114	Australian Collectors, Co.	Australia	2003-05-31	7565.08
114	Australian Collectors, Co.	Australia	2004-03-10	44894.74
119	La Rochelle Gifts	France	2004-11-14	19501.82
119	La Rochelle Gifts	France	2004-08-08	47924.19
119	La Rochelle Gifts	France	2005-02-22	49523.67
121	Baane Mini Imports	Norway	2003-02-16	50218.95
121	Baane Mini Imports	Norway	2003-10-28	1491.38
121	Baane Mini Imports	Norway	2004-11-04	17876.32
121	Baane Mini Imports	Norway	2004-11-28	34639.14
124	Mini Gifts Distributors Ltd.	USA	2005-03-05	101244.59
124	Mini Gifts Distributors Ltd.	USA	2004-08-28	85410.07
124	Mini Gifts Distributors Ltd.	USA	2003-04-11	11044.30
124	Mini Gifts Distributors Ltd.	USA	2005-04-16	83598.04
124	Mini Gifts Distributors Ltd.	USA	2004-12-27	47142.70
124	Mini Gifts Distributors Ltd.	USA	2004-11-02	55639.66
124	Mini Gifts Distributors Ltd.	USA	2003-08-15	111654.40
124	Mini Gifts Distributors Ltd.	USA	2004-03-26	43369.30
124	Mini Gifts Distributors Ltd.	USA	2003-11-25	45084.38
128	Blauer See Auto, Co.	Germany	2003-01-28	10549.01
128	Blauer See Auto, Co.	Germany	2003-10-18	24101.81
128	Blauer See Auto, Co.	Germany	2004-03-24	33820.62
128	Blauer See Auto, Co.	Germany	2004-11-18	7466.32
129	Mini Wheels Co.	USA	2004-12-08	26248.78
129	Mini Wheels Co.	USA	2003-12-11	23923.93
129	Mini Wheels Co.	USA	2003-04-09	16537.85
131	Land of Toys Inc.	USA	2003-03-12	22292.62
131	Land of Toys Inc.	USA	2004-12-02	50025.35
131	Land of Toys Inc.	USA	2004-09-11	35321.97

Figure 30: SQL Query Execution Result

```
mydb=# SELECT
c."country",
COUNT(*) AS "customerCount"
FROM customers AS c
GROUP BY c."country"
ORDER BY "customerCount" DESC, c."country";
```

country	customerCount
USA	36
Germany	13
France	12
Spain	7
Australia	5
UK	5
Italy	4
New Zealand	4
Canada	3
Finland	3
Singapore	3
Switzerland	3
Austria	2
Belgium	2
Denmark	2
Ireland	2
Japan	2
Norway	2
Portugal	2
Sweden	2
Hong Kong	1
Israel	1
Netherlands	1
Norway	1
Philippines	1
Poland	1
Russia	1
South Africa	1

(28 rows)

Figure 31: SQL Query Execution Result

customerNumber	customerName	totalPayments
141	Euro+ Shopping Channel	715738.98
124	Mini Gifts Distributors Ltd.	584188.24
114	Australian Collectors, Co.	180585.07
151	Muscle Machine Inc	177913.95
148	Dragon Souvenirs, Ltd.	156251.03
323	Down Under Souvenirs, Inc	154622.08
187	AV Stores, Co.	148410.09
276	Anna's Decorations, Ltd	137034.22
321	Corporate Gift Ideas Co.	132340.78
146	Saveley & Henriot, Co.	130305.35
278	Rovelli Gifts	127529.69
353	Reims Collectables	126983.19
119	La Rochelle Gifts	116949.68
363	Online Diecast Creations Co.	116449.29
496	Kelly's Gift Shop	114497.19
458	Corrida Auto Replicas, Ltd	112440.09
298	Vida Sport, Ltd	108777.92
131	Land of Toys Inc.	107639.94
145	Danish Wholesale Imports	107446.50
398	Tokyo Collectables, Ltd	105548.73

Figure 32: SQL Query Execution Result

```
mydb=# SELECT
  o."status",
  COUNT(*) AS "orderCount"
FROM orders AS o
GROUP BY o."status"
ORDER BY "orderCount" DESC, o."status";
```

status	orderCount
Shipped	303
Cancelled	6
In Process	6
On Hold	4
Resolved	4
Disputed	3

(6 rows)

Figure 33: SQL Query Execution Result

```
mydb=# SELECT
  p."productLine",
  COUNT(*) AS "productCount"
FROM products AS p
GROUP BY p."productLine"
ORDER BY "productCount" DESC, p."productLine";
```

productLine	productCount
Classic Cars	38
Vintage Cars	24
Motorcycles	13
Planes	12
Trucks and Buses	11
Ships	9
Trains	3

(7 rows)

Figure 34: SQL Query Execution Result

```
mydb=# SELECT
  o."officeCode",
  o."city",
  COUNT(e."employeeNumber") AS "employeeCount"
FROM offices AS o
LEFT JOIN employees AS e
  ON e."officeCode" = o."officeCode"
GROUP BY o."officeCode", o."city"
ORDER BY "employeeCount" DESC, o."officeCode";
 officeCode |      city      | employeeCount
-----+-----+-----
1          | San Francisco |             6
4          | Paris         |             5
6          | Sydney        |             4
2          | Boston        |             2
3          | NYC           |             2
5          | Tokyo         |             2
7          | London        |             2
(7 rows)
```

Figure 35: SQL Query Execution Result

```
mydb=# SELECT
  p."productVendor",
  SUM(p."quantityInStock") AS "totalStock"
FROM products AS p
GROUP BY p."productVendor"
ORDER BY "totalStock" DESC, p."productVendor";
 productVendor | totalStock
-----+-----
Gearbox Collectibles |        60495
Min Lin Diecast      |        50089
Classic Metal Creations |       45408
Welly Diecast Productions |       45095
Exoto Designs        |       44166
Motor City Art Classics |       43105
Second Gear Diecast   |       42865
Studio M Art Models   |       42253
Carousel DieCast Legends |      40805
Unimax Art Galleries  |       38191
Highway 66 Mini Classics |      37520
Red Start Diecast     |       35046
Autoart Studio Design |       30093
(13 rows)
```

Figure 36: SQL Query Execution Result

```
mydb=# SELECT
  p."productLine",
  AVG(p."buyPrice") AS "avgBuyPrice"
FROM products AS p
GROUP BY p."productLine"
ORDER BY "avgBuyPrice" DESC, p."productLine";
 productLine | avgBuyPrice
-----+-----
Classic Cars | 64.4463157894736842
Trucks and Buses | 56.32909090909091
Motorcycles | 50.6853846153846154
Planes | 49.629166666666667
Ships | 47.007777777777778
Vintage Cars | 46.066250000000000
Trains | 43.923333333333333
(7 rows)
```

Figure 37: SQL Query Execution Result

customerNumber	customerName	orderCount
141	Euro+ Shopping Channel	26
124	Mini Gifts Distributors Ltd.	17
114	Australian Collectors, Co.	5
145	Danish Wholesale Imports	5
148	Dragon Souvenirs, Ltd.	5
323	Down Under Souvenirs, Inc	5
353	Reims Collectables	5
119	La Rochelle Gifts	4
121	Baane Mini Imports	4
128	Blauer See Auto, Co.	4
131	Land of Toys Inc.	4
144	Volvo Model Replicas, Co	4
151	Muscle Machine Inc	4
157	Diecast Classics Inc.	4
161	Technics Stores Inc.	4
166	Handji Gifts& Co	4
276	Anna's Decorations, Ltd	4
282	Souvenirs And Things Co.	4
321	Corporate Gift Ideas Co.	4
381	Royale Belge	4
382	Salzburg Collectables	4
398	Tokyo Collectables, Ltd	4
450	The Sharp Gifts Warehouse	4
496	Kelly's Gift Shop	4
103	Atelier graphique	3
112	Signal Gift Stores	3
129	Mini Wheels Co.	3
146	Saveley & Henriot, Co.	3
167	Herkku Gifts	3
172	La Corne Dabondance, Co.	3

Figure 38: SQL Query Execution Result

```
mydb=# SELECT
  p."productLine",
  MAX(p."MSRP") AS "maxMsrp"
FROM products AS p
GROUP BY p."productLine"
ORDER BY "maxMsrp" DESC, p."productLine";
 productLine | maxMsrp
-----+-----
Classic Cars | 214.30
Motorcycles  | 193.66
Vintage Cars | 170.00
Planes       | 157.69
Trucks and Buses | 136.67
Ships        | 122.89
Trains       | 100.84
(7 rows)
```

Figure 39: SQL Query Execution Result

```
mydb=# SELECT
  COUNT(*) AS "totalCustomers"
FROM customers;
totalCustomers
-----
122
(1 row)
```

Figure 40: SQL Query Execution Result

```
mydb=# SELECT
COUNT(*) AS "totalProducts"
FROM products;
totalProducts
-----
110
(1 row)
```

Figure 43: SQL Query Execution Result

```
mydb=# SELECT
SUM(p."amount") AS "totalPaymentRevenue"
FROM payments AS p;
totalPaymentRevenue
-----
8853839.23
(1 row)
```

Figure 44: SQL Query Execution Result

```
mydb=# SELECT
AVG(p."buyPrice") AS "avgBuyPrice"
FROM products AS p;
avgBuyPrice
-----
54.39518181818182
(1 row)
```

Figure 45: SQL Query Execution Result

```
mydb=# SELECT
MAX(p."amount") AS "maxPaymentAmount"
FROM payments AS p;
maxPaymentAmount
-----
120166.58
(1 row)
```

Figure 46: SQL Query Execution Result

```
mydb=# SELECT
MIN(p."amount") AS "minPaymentAmount"
FROM payments AS p;
minPaymentAmount
-----
615.45
(1 row)
```

Figure 47: SQL Query Execution Result

```
mydb=# SELECT
      COUNT(*) AS "totalOrders"
FROM orders;
totalOrders
-----
          326
(1 row)
```

Figure 48: SQL Query Execution Result

```
mydb=# SELECT
      SUM(p."quantityInStock") AS "totalQuantityInStock"
FROM products AS p;
totalQuantityInStock
-----
          555131
(1 row)
```

Figure 49: SQL Query Execution Result

```
mydb=# SELECT
      AVG(p."MSRP") AS "avgMsrp"
FROM products AS p;
avgMsrp
-----
100.43872727272727
(1 row)
```

Figure 50: SQL Query Execution Result

```
mydb=# SELECT
      COUNT(*) AS "totalEmployees"
FROM employees;
totalEmployees
-----
           23
(1 row)
```

Part 2: Evaluation Strategy for Text-to-SQL Agents

Goals:

- Verify SQL correctness beyond syntax validity
- Ensure returned results accurately reflect user intent
- Measure robustness, retry behavior, efficiency, and safety

Evaluation Methods:

1. Exact SQL Match

- Compare generated SQL with normalized gold queries.

2. Execution Accuracy

- Execute both gold and generated queries on the same database.
- Compare outputs using schema and set equality checks.

3. Component-Level Validation

- Validate selected tables, joins, filters, and aggregation logic.

4. Error Handling and Retry Analysis

- Measure retry count and convergence success after failures.

5. Natural Language Answer Quality

- Evaluate correctness, clarity, and completeness of generated responses.

6. Efficiency and Safety

- Track execution latency and prevent unsafe operations.

Metrics:

- Exact Match Rate
- Execution Accuracy
- Component Accuracy
- Retry Success Rate
- Average Query Latency
- Safety Rate

Recommended Procedure:

1. Execute gold queries and cache outputs.
2. Run generated queries against the same database.
3. Compare outputs and schemas.
4. Log retries, execution times, and failures.
5. Generate final benchmark scores and summaries.

Appendix: PostgreSQL + Docker Setup

Start PostgreSQL using Docker:

```
docker run --name mydb -e POSTGRES_PASSWORD=postgres -e POSTGRES_DB=mydb -p 5432:5432 -d postgres:15
```

Seed the Database:

```
Get-Content .\seed.sql | docker exec -i mydb psql -U postgres -d mydb
```


Connect with psql:

```
psql -h localhost -U postgres -d mydb
```

Can deal with Large Result:

```
SELECT COUNT(*) FROM (<your_query_here>) AS q;
```

```
SELECT * FROM (<your_query_here>) AS q LIMIT 5;
```

Conclusion

The benchmark submission demonstrates a broad range of SQL operations performed on the classicmodels database along with a structured evaluation framework for Text-to-SQL systems. The included screenshots validate successful query execution and result generation.